

A Repaired Cycle-Monotonicity Lower Bound for Nonseparable SOS Combinatorial Auctions

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Abstract

This note revisits a proposed lower-bound construction for the open problem of Eden, Feldman, Fiat, Goldner, and Karlin on whether separability is necessary for constant welfare approximation in combinatorial auctions with interdependent SOS valuations. The proposed construction is not valid for the formal model in [1], because it assumes that every item must be allocated. In the model of [1], allocated bundles need only be disjoint; items may be left unallocated.

I give a repaired construction in the actual non-total-allocation model. The repair is deliberately conservative: it proves a valid lower bound approaching 2 for universally ex-post IC-IR randomized mechanisms with two agents, multidimensional bundle signals, and nonseparable SOS valuations. Thus it is *partial progress only*; it is not a solution of the open problem and does not show that constant approximation is impossible.

The note also proves a barrier explaining why the original proof strategy cannot be upgraded directly to an unbounded lower bound. In any two-cycle exact-allocation obstruction of this kind, either dropping the cyclic agent from the optimum or giving that agent her tempting bundle already yields at least half of the optimum welfare. Therefore this proof template cannot rule out 2-approximation in the no-disposal-of-agents / disposal-of-items model.

1 Status and relation to the original attempt

The relevant combinatorial-auction model in [1] is not a total-partition model. A deterministic mechanism chooses, for every agent i and bundle T , an allocation indicator x_{iT} and payments, with the constraints that each agent receives at most one bundle and allocated bundles are disjoint. There is no constraint requiring the union of the allocated bundles to be all items.

The original construction used the following idea. At profile r , agent 2 has value 1 for item r , and agent 1 is forced by total partitioning to receive $[q] \setminus \{r\}$. Across two profiles $r \neq h$, agent 1 prefers receiving $[q] \setminus \{h\}$ at type r and receiving $[q] \setminus \{r\}$ at type h , so exact efficiency at both profiles violates a two-cycle monotonicity constraint.

In the actual model, this exact argument collapses: the mechanism can give item r to agent 2 and leave all other items unallocated. Agent 1 then receives nothing, so the cycle through agent 1 disappears. The repair below adds welfare to the bundle received by agent 1 at the intended allocation. This makes disposal costly, but only enough to obtain a factor approaching 2.

2 Model and cycle monotonicity

There are n agents and q items. In the multidimensional-signal combinatorial model, agent i has a signal s_{iT} for every bundle $T \subseteq [q]$. The value of agent i for bundle T is

$$v_{iT}(s_T), \quad s_T = (s_{1T}, \dots, s_{nT}).$$

We use the formal feasibility constraints of [1]: each agent is allocated at most one bundle, and allocated bundles are pairwise disjoint. Items not allocated to any agent are allowed.

A deterministic mechanism is ex-post IC if, for every true signal profile and for every agent, truthful reporting maximizes utility assuming all other agents report truthfully. A randomized mechanism is universally ex-post IC-IR if it is a distribution over deterministic ex-post IC-IR mechanisms.

We use only the following standard consequence of ex-post implementability.

Lemma 1 (Two-cycle monotonicity). *Fix an agent i and fix the other agents' reports. Let a, b be two possible reports of agent i . Suppose a deterministic ex-post IC allocation rule assigns bundles $X_i(a)$ and $X_i(b)$ to agent i at these two reports. Then*

$$v_{i,X_i(a)}(a) - v_{i,X_i(b)}(a) + v_{i,X_i(b)}(b) - v_{i,X_i(a)}(b) \geq 0.$$

Proof. Let $p_i(a)$ and $p_i(b)$ be the corresponding payments, with the other reports fixed. Truthfulness for type a gives

$$v_{i,X_i(a)}(a) - p_i(a) \geq v_{i,X_i(b)}(a) - p_i(b),$$

and truthfulness for type b gives

$$v_{i,X_i(b)}(b) - p_i(b) \geq v_{i,X_i(a)}(b) - p_i(a).$$

Adding the two inequalities cancels payments and gives the claim. $\square$

For twice continuously differentiable two-signal valuations, the SOS condition is implied by non-positive mixed partial derivative. Thus, for $f : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$, if $\partial^2 f / \partial a \partial b \leq 0$ and f is coordinatewise increasing, then increasing one signal has a weakly smaller marginal effect when the other signal is larger.

3 A repaired hard family

Fix an integer $q \geq 3$, a parameter $0 < \delta < 1$, and an arbitrary small parameter $0 < \eta < 1$. Let the item set be $[q] = \{1, \dots, q\}$. For each $h \in [q]$, write

$$C_h = [q] \setminus \{h\}.$$

Thus C_h is the complement of item h .

The construction has two agents. The designated signal profiles are $s^1, \dots, s^q$. Agent 2's signals are always zero:

$$s_{2T}^r = 0 \quad \text{for all } r \in [q] \text{ and } T \subseteq [q].$$

Agent 1's signals at profile r are as follows:

$$s_{1T}^r = \begin{cases} 1, & T = \{r\}, \\ 1, & T = C_r, \\ 1 + \delta, & T = C_h \text{ for some } h \neq r, \\ 0, & \text{otherwise.} \end{cases}$$

The assumption $q \geq 3$ ensures that singleton bundles and complement bundles are distinct.

Define the smooth function

$$G(a, b) = a + b - (1 - e^{-a})(1 - e^{-b}), \quad a, b \in \mathbb{R}_+.$$

The valuations are:

$$v_{1T}(a, b) = \begin{cases} a, & T = C_h \text{ for some } h \in [q], \\ \eta a, & \text{otherwise,} \end{cases}$$

and

$$v_{2T}(a, b) = \begin{cases} G(a, b), & |T| = 1, \\ \eta b, & |T| \neq 1. \end{cases}$$

Here $a = s_{1T}$ and $b = s_{2T}$.

Lemma 2 (SOS and nonseparability). *The above valuations are weakly increasing in all signal coordinates, strictly increasing in the agent's own coordinate, and SOS. The instance is not separable-SOS.*

Proof. For agent 1, every valuation is linear in agent 1's own signal and independent of agent 2's signal. The coefficient of agent 1's signal is either 1 or $\eta > 0$, so the valuation is strictly increasing in the own coordinate, weakly increasing in the other coordinate, and has zero mixed partial.

For agent 2 and non-singleton bundles, $v_{2T}(a, b) = \eta b$, which is linear, weakly increasing in a , and strictly increasing in agent 2's own coordinate b .

For singleton bundles,

$$\frac{\partial G}{\partial a} = 1 - e^{-a}(1 - e^{-b}), \quad \frac{\partial G}{\partial b} = 1 - e^{-b}(1 - e^{-a}).$$

Both derivatives are positive for all $a, b \geq 0$. Moreover,

$$\frac{\partial^2 G}{\partial a \partial b} = -e^{-a-b} < 0.$$

Thus G is SOS. The same strictly negative mixed partial also shows that G cannot be written as a sum of a function of a and a function of b . Therefore the valuation $v_{2, \{j\}}$ is not separable, and the instance is not separable-SOS. $\square$

For each profile r , define the intended allocation A^r by

$$A_1^r = C_r, \quad A_2^r = \{r\}.$$

This allocation leaves no items unallocated, but the mechanism is not required to do so.

Lemma 3 (Welfare at designated profiles). *At signal profile s^r , allocation A^r is the unique welfare-maximizing allocation and has welfare 2. Every other feasible allocation has welfare at most $1 + \delta$.*

Proof. At profile s^r , agent 1's value for C_r is 1, and agent 2's value for singleton $\{r\}$ is

$$G(1, 0) = 1.$$

Thus $\text{SW}(A^r; s^r) = 2$.

Now consider any feasible allocation $(T_1, T_2) \neq (C_r, \{r\})$. If $T_2 = \{r\}$, then agent 2 contributes 1. Since T_1 must be disjoint from $\{r\}$, the only complement bundle available to agent 1 is C_r . If

$T_1 = C_r$, the allocation is exactly A^r , contrary to assumption. Hence, in this case, agent 1 receives a non-complement bundle whose signal is zero at profile s^r , so the total welfare is at most 1.

If $T_2 \neq \{r\}$, then agent 2 contributes zero: wrong singletons have signal pair $(0, 0)$, and non-singleton bundles have $s_{2T} = 0$. Agent 1 receives at most one bundle, and her maximum possible value at s^r is $1 + \delta$, attained by the complement bundles C_h with $h \neq r$. Therefore the total welfare is at most $1 + \delta$.

Since $\delta < 1$, every non- A^r allocation has welfare strictly below 2, so A^r is the unique optimum. $\square$

Lemma 4 (A deterministic rule is exact at at most one designated profile). *Let x be any deterministic ex-post IC allocation rule. Then there is at most one $r \in [q]$ such that $x(s^r) = A^r$.*

Proof. Suppose not. Then for some distinct $r, h \in [q]$, the rule chooses A^r at s^r and A^h at s^h . Agent 2's report is the same at all designated profiles, namely the all-zero report, so we may apply Lemma 1 to agent 1 between the two reports s_1^r and s_1^h .

At true profile s^r , agent 1's value for the bundle she receives under A^r is

$$v_{1,C_r}(s_{C_r}^r) = 1,$$

while her value for the bundle she would receive under A^h is

$$v_{1,C_h}(s_{C_h}^r) = 1 + \delta,$$

because $h \neq r$. Similarly,

$$v_{1,C_h}(s_{C_h}^h) = 1, \quad v_{1,C_r}(s_{C_r}^h) = 1 + \delta.$$

The two-cycle weight is therefore

$$(1 - (1 + \delta)) + (1 - (1 + \delta)) = -2\delta < 0,$$

contradicting ex-post IC. $\square$

Theorem 1 (A valid lower bound approaching 2). *For every $q \geq 3$ and every $0 < \delta < 1$, there is a two-agent, q -item, multidimensional-signal, nonseparable-SOS instance in the formal combinatorial-auction model of [1] such that every universally ex-post IC-IR randomized mechanism has worst-case approximation ratio at least*

$$\rho(q, \delta) = \frac{2}{1 + \delta + (1 - \delta)/q}.$$

Consequently, for every $\gamma > 0$, there is such an instance on which every universally ex-post IC-IR randomized mechanism has approximation ratio greater than $2 - \gamma$.

Proof. Let $\mathcal{M}$ be a universally ex-post IC-IR randomized mechanism. Write it as a distribution over deterministic ex-post IC-IR mechanisms. For each $r \in [q]$, let

$$p_r = \Pr[\mathcal{M}(s^r) = A^r].$$

By Lemma 4, each deterministic rule in the support can contribute to the event $\{\mathcal{M}(s^r) = A^r\}$ for at most one value of r . Hence

$$\sum_{r=1}^q p_r \leq 1.$$

Choose r^* with $p_{r^*} \leq 1/q$.

At s^{r*} , the optimum welfare is 2 by Lemma 3. If $\mathcal{M}$ chooses A^{r*} , it obtains welfare 2; otherwise, by Lemma 3, it obtains welfare at most $1 + \delta$. Therefore

$$\begin{aligned}\mathbb{E}[\text{SW}(\mathcal{M}(s^{r*}); s^{r*})] &\leq p_{r*} \cdot 2 + (1 - p_{r*})(1 + \delta) \\ &= 1 + \delta + p_{r*}(1 - \delta) \\ &\leq 1 + \delta + \frac{1 - \delta}{q}.\end{aligned}$$

Dividing the optimum value 2 by this upper bound gives the claimed ratio.

Finally, as $q \rightarrow \infty$ and $\delta \rightarrow 0$, the displayed ratio tends to 2. Thus for every $\gamma > 0$ one can choose q large and δ small enough so that $\rho(q, \delta) > 2 - \gamma$. $\square$

4 Why this is not a solution of the open problem

Theorem 1 is a genuine lower bound in the actual model, but it is only a constant lower bound. It is therefore compatible with the existence of a constant-factor universally ex-post IC-IR mechanism for all nonseparable SOS valuations. It does not prove that separability is necessary for constant approximation.

Indeed, on the designated profiles of the construction, there is an obvious truthful disposal mechanism achieving welfare exactly 1, i.e., a factor 2 on these profiles: use agent 1's reported singleton signals to choose a singleton $\{j\}$ for agent 2, allocate nothing to agent 1, and charge zero payments. Agent 1 receives nothing and pays zero, so her utility is independent of her report. Agent 2's allocation is independent of agent 2's report. Thus the repaired construction cannot possibly prove a lower bound above 2 by analyzing only these profiles.

The next proposition gives a more structural explanation.

Proposition 1 (Half-optimal escape from a two-cycle exactness argument). *Consider any combinatorial-auction instance in the non-total-allocation model. Fix an agent i and two reports a, b of that agent, with all other reports fixed. Let A^a and A^b be two feasible allocations. Suppose that at type a , agent i strictly prefers her bundle in A^b to her bundle in A^a :*

$$v_i(A_i^b; a) > v_i(A_i^a; a).$$

Let $W = \text{SW}(A^a; a)$, and suppose A^a is an intended optimum at type a . Then at type a there is a feasible allocation different from the exact allocation A^a whose welfare is at least $W/2$.

Proof. Write

$$x = v_i(A_i^a; a), \quad y = v_i(A_i^b; a),$$

so $y > x$. First, remove agent i 's bundle from A^a and keep all other agents' bundles as in A^a . Because the model allows items to remain unallocated, this is feasible and has welfare $W - x$.

Second, allocate bundle A_i^b to agent i alone and allocate nothing to everyone else. This is also feasible and has welfare y .

If $x \leq W/2$, the first allocation has welfare $W - x \geq W/2$. If $x > W/2$, then the second allocation has welfare $y > x > W/2$. Hence some non-exact feasible allocation has welfare at least $W/2$. $\square$

Corollary 1 (Template limitation). *No lower-bound proof that relies only on showing that exact optimal allocations along a two-cycle are not simultaneously implementable can certify that all non-exact allocations have welfare below half of optimum. Therefore this exact-allocation two-cycle template cannot by itself yield an unbounded approximation lower bound in the non-total-allocation model.*

5 Conclusion

The original total-partition proof cannot be repaired into an unbounded lower bound by a local patch. A valid repair is possible, but it yields only a $2 - o(1)$ lower bound. The construction shows that nonseparable SOS valuations can reproduce the standard cycle-monotonicity tension in the exact formal model of [1]; however, the ability to leave items unallocated gives a half-optimal escape route from the simple two-cycle strategy.

Thus the present note should be read as partial progress and a diagnostic barrier, not as a resolution of the separability open problem. Any proof that separability is necessary for constant approximation would need a mechanism-obstruction idea that controls all high-welfare allocations, not merely the exact optimum allocations involved in a two-cycle.

References

- [1] Alon Eden, Michal Feldman, Amos Fiat, Kira Goldner, and Anna R. Karlin. *Combinatorial Auctions with Interdependent Valuations: SOS to the Rescue*. Proceedings of the 20th ACM Conference on Economics and Computation, 2019. arXiv:1903.08384.
- [2] Jean-Charles Rochet. A necessary and sufficient condition for rationalizability in a quasi-linear context. *Journal of Mathematical Economics*, 16(2):191–200, 1987.